

Jaccard Distance/ Dissimilarity Calculations

The Jaccard calculations apply to asymmetric data with binary attributes (0,1) and varying attribute importance (Aggarwal and Reddy, 2014). To calculate the Jaccard distance for the table below, I will convert the attributes into binary values based on the category representations. The presence/positive state will then be encoded as 1 and absence/negative state as 0.

Name	Gender	Fever	Cough	Test-1	Test-2	Test-3	Test-4
Jack	M	Y	N	P	N	N	A
Mary	F	Y	N	P	A	P	N
Jim	M	Y	P	N	N	N	A

Conversion decisions:

- **Gender:** Since both genders are equally important, gender will not be encoded and will be excluded from the calculations; the focus will be on symptoms and test outcomes.
- **Fever:** Y = 1
- **Cough:** P (productive, wet) indicates an infection such as flu, bronchitis, or pneumonia vs N (non-productive, dry) suggests a regular cough (National Library of Medicine, n.d.)
 - P = 1
 - N = 0
- **Test:** The test results, negative and normal vs Positive or abnormal (Villines, Z, 2024)
 - N - negative = 0
 - P - positive = 1
 - A = abnormal = 1

Conversion decisions:

Name	Gender	Fever	Cough	Test-1	Test-2	Test-3	Test-4
Jack	M	1	0	1	0	0	1
Mary	F	1	0	1	1	1	0
Jim	M	1	1	0	0	0	1

Jaccard distance calculation

To calculate the distance, I first assess Jaccard similarity (Han, Kamber and Pei; 2012), which accounts for shared presence (1,1) relative to all cases where at least one observation has the attribute, while ignoring shared absence (0,0).

J similarity formula / coefficient = $F(1,1) / [F(1,1) + F(1,0) + F(0,1)]$

1. (Jack, Mary)

Name	Fever	Cough	Test-1	Test-2	Test-3	Test-4
Jack	1	0	1	0	0	1
Mary	1	0	1	1	1	0

Jaccard similarity: $2 / (2 + 3 + 1) = 2 / 6 \Rightarrow 33,33\%$

Jacquard distance = 66.67%

2. (Jack, Jim)

Name	Fever	Cough	Test-1	Test-2	Test-3	Test-4
Jack	1	0	1	0	0	1
Jim	1	1	0	0	0	1

Jaccard similarity: $2 / (1 + 1 + 2) = 2 / 4 \Rightarrow 50\%$

Jacquard distance = 50%

3. (Mary, Jim)

Name	Fever	Cough	Test-1	Test-2	Test-3	Test-4
Mary	1	0	1	1	1	0
Jim	1	1	0	0	0	1

Jaccard similarity: $1 / (1 + 4 + 2) = 1 / 7 \Rightarrow 14.29\%$

Jacquard distance = 85.71%

Summary Jaccard calculations:

Pairs	Jaccard Similarity	Jaccard Distance
Jack, Mary	33.33%	66.67%
Jack, Jim	50.00%	50.00%
Mary, Jim	14.29%	85.71%

Jack and Jim share the highest similarity and a similar sickness pattern with the lowest dissimilarity measure. At the same time, Mary and Jim are unlikely to have the same disease, given their highest distance.

References

Aggarwal, C.C. and Reddy, C.K., eds. (2014) *Data Clustering: Algorithms and Applications*. 1st edn. Vol. 31. Boca Raton: Chapman & Hall.

Han, J., Kamber, M. and Pei, J. (2012) *Data Mining: Concepts and Techniques*. 3rd edn. Waltham, MA: Morgan Kaufmann.

National Library of Medicine (n.d.) *How to understand your lab results*. MedlinePlus. Available at: <https://medlineplus.gov/lab-tests/how-to-understand-your-lab-results/> (Accessed: 26 August 2026).

Villines, Z. (2024) 'What to know about blood glucose levels', *Medical News Today*. Updated 14 August. Available at: <https://www.medicalnewstoday.com/articles/327142> (Accessed: 26 August 2026)